Seeing the Unseen: Multi-Objective Rescue of the Fashion Long Tail

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Abstract

Fashion recommendation systems disproportionately promote popular items, burying the vast majority of a retailer's catalog — the *long tail*. We present a three-stage framework that progressively addresses this problem on the H&M Personalized Fashion dataset (∼31M transactions, ∼105K product images). **Stage 1** establishes a deliberately biased baseline (the "Villain"), a SASRec model augmented with a learnable popularity-bias vector. **Stage 2** introduces the "Hero", a Behavior Sequence Transformer (BST) fusing item-ID and ResNet50 visual embeddings, trained with a combined cross-entropy and InfoNCE contrastive loss with attribute-aware hard-negative mining. **Stage 3** adds a differentiable discovery loss and conducts a multi-objective Pareto sweep over its weight λ_{disc}. At the Pareto-optimal operating point (λ=0.3), catalog coverage rises by +7.6 percentage points (pp) and tail-item recommendation rate triples (2.2%→6.6%) with zero nDCG@12 sacrifice relative to the base Hero. A visual ablation further reveals that ResNet50 embeddings improve cold-start item ranking by over 4,500 positions.

1 Introduction

Global fashion platforms such as H&M manage catalogs exceeding 100,000 items, yet standard recommender systems overwhelmingly surface a small set of popular products. This *popularity bias* creates a self-reinforcing feedback loop: popular items receive more exposure, generating more interactions, which further increases their ranking. Meanwhile, roughly 85% of the catalog — the *long tail* — remains invisible to users despite containing products that may better match their individual style.

This project addresses the long-tail gap through a three-act narrative:

1. **The Villain** — a position-aware SASRec baseline that is intentionally blind to product images and amplifies popularity bias via a learnable `pop_bias` vector.

2. **The Hero** — a Behavior Sequence Transformer (BST) that fuses item-ID embeddings with ResNet50 visual embeddings and is trained with a combined cross-entropy and InfoNCE contrastive loss.

3. **The Brain** — a multi-objective extension that adds a differentiable discovery loss, enabling a tuneable trade-off between ranking accuracy and catalog exploration.

We evaluate all models on a multi-objective metric suite comprising nDCG@12, MRR, catalog coverage, and tail-item recommendation rate.

2 Related Work

Sequential recommendation. Self-Attentive Sequential Recommendation (SASRec) [1] models user purchase sequences via causal Transformer attention and has become a standard baseline in the field.

Multimodal recommendation. Behavior Sequence Transformers (BST) [2] incorporate side information into the Transformer encoder. Visual features from CNNs such as ResNet50 [3] have been shown to improve cold-start and long-tail item representation [4].

Long-tail & fairness. Prior work addresses popularity bias through inverse-propensity weighting [5], causal debiasing [6], and multi-objective re-ranking [7]. Our approach differs by integrating a *differentiable* discovery loss directly into the training objective, avoiding post-hoc re-ranking.

Contrastive learning for recommendations. InfoNCE-based contrastive objectives have been applied to create structured embedding spaces [8], and attribute-aware hard-negative mining sharpens the contrastive signal beyond random sampling.

3 Dataset & Preprocessing

We use the **H&M Personalized Fashion Recommendations** dataset, comprising ∼31M transactions from ∼1.37M customers across ∼105K articles with product images.

3.1 Sampling & Temporal Pruning

To enable local GPU training (NVIDIA RTX 5070 Ti, 12 GB VRAM), we apply stratified sampling:

- Retain the most recent 12 weeks of transactions (temporal pruning).
- Sample 15% of users, stratified by the popularity bin of their dominant purchase category, preserving the head/torso/tail distribution.
- Filter users with ≥ 5 interactions.

The resulting dataset contains **430,879 transactions**, **36,234 users**, and **26,932 articles**. Items with fewer than 10 total purchases are labelled as *long-tail* (∼85% of the catalog).

3.2 Data Split

We adopt a **leave-one-out temporal split**: the last item in each user's sequence is held out for testing, the second-to-last for validation, and the remainder for training. Sliding-window augmentation is applied to the training prefix.

3.3 Visual Embedding Extraction

All ∼105K product images are passed through a frozen **ResNet50** (pre-trained on ImageNet), yielding 2048-dimensional feature vectors per article. These are stored as memory-mapped `.npy` files (∼0.8 GB).

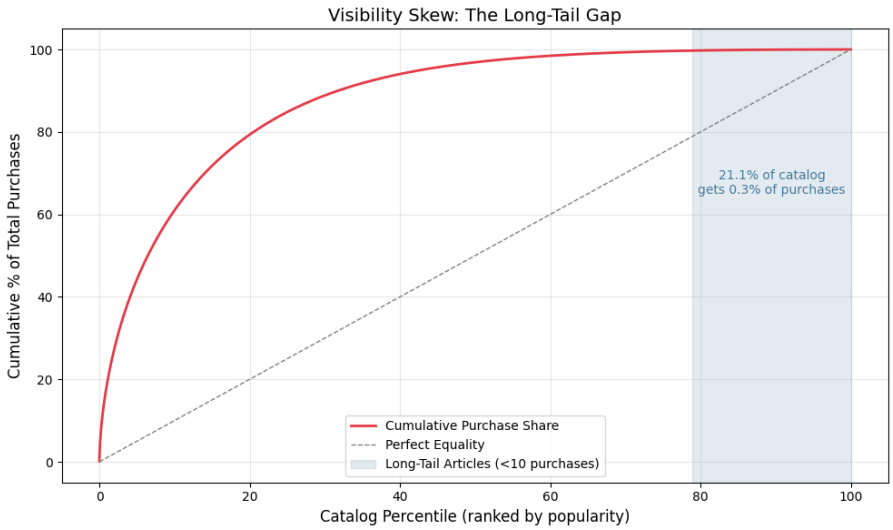


Figure 1: Visibility skew in the H&M catalog. The vast majority of articles occupy the long tail with minimal transaction volume, while a small fraction of head items dominate.

4 Methodology

4.1 Act 1: The Villain (SASRec + Popularity Bias)

The Villain is a **SASRec**-based sequential recommender with an intentionally added popularity amplifier.

Embedding layer. Item IDs and position indices are mapped to D=128-dimensional embeddings via learned lookup tables, then summed and layer-normalised:

$$\mathbf{x} = \text{LayerNorm}(\mathbf{E}_{\text{item}}[\mathbf{s}] + \mathbf{E}_{\text{pos}}[\mathbf{p}]) \tag{1}$$

where $\mathbf{s} \in \mathbb{Z}^{B \times S}$ is the padded item sequence and $\mathbf{p}$ the position indices.

Transformer encoder. A 3-layer causal Transformer encoder with 4 attention heads processes the sequence. A causal mask ensures each position attends only to itself and earlier positions.

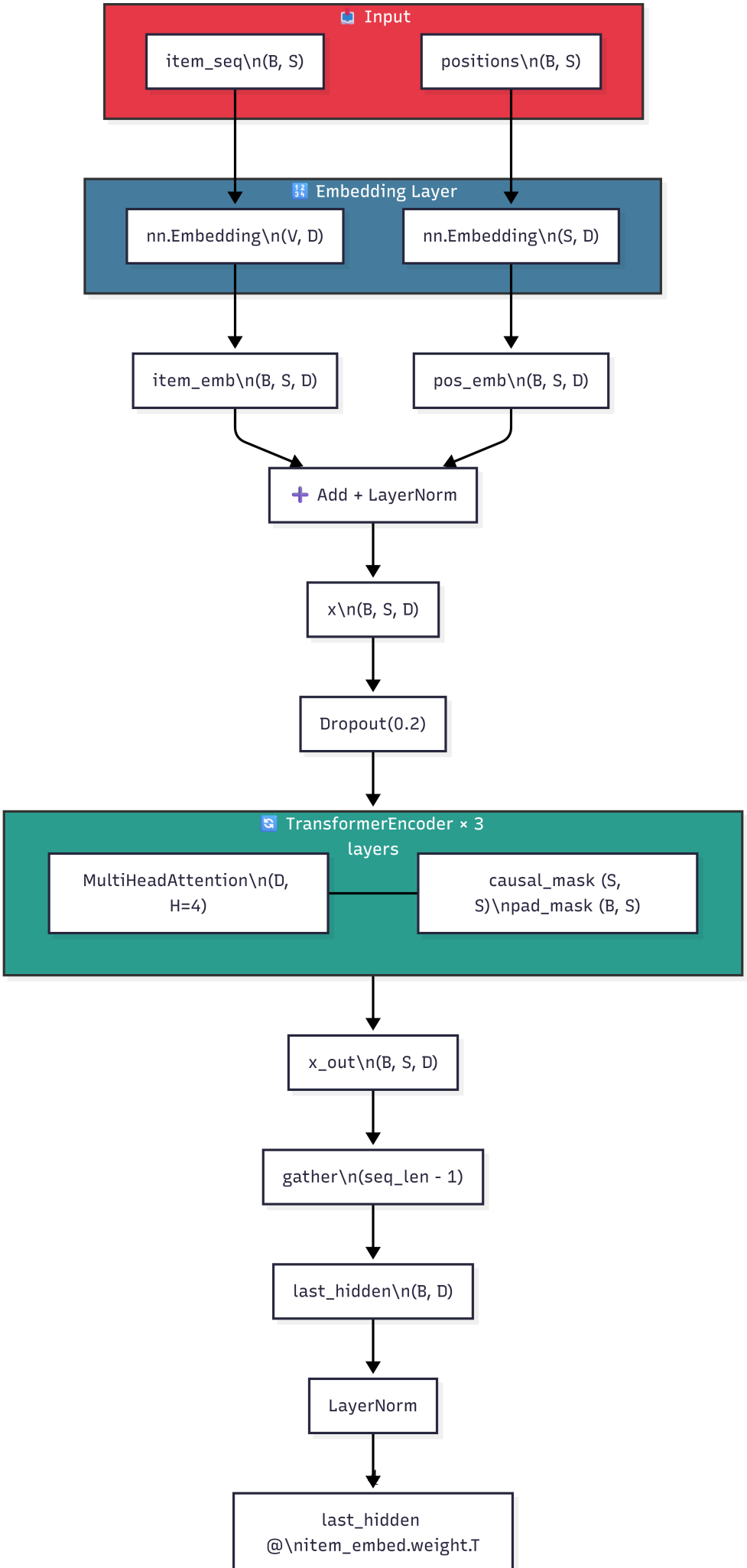
Prediction head. The hidden state at the last valid position is projected against the item embedding matrix:

$$\mathbf{z} = \mathbf{h}_{\text{last}} \cdot \mathbf{E}_{\text{item}}^{\top} \in \mathbb{R}^{V} \tag{2}$$

Popularity bias. A learnable vector $\mathbf{b} \in \mathbb{R}^{V}$ (initialised to ones) multiplies the logits element-wise: $\hat{\mathbf{z}} = \mathbf{z} \odot \mathbf{b}$. This amplifies scores for popular items. After training, $\mathbf{b}$ values range from 0.83 to 2.08, confirming the model actively exploits popularity.

Training. Cross-entropy loss over the full V=26,933 vocabulary, AdamW optimiser (lr $= 10^{-3}$, weight decay 0.01), ReduceLROnPlateau scheduler, batch size 256, early stopping with patience 10. The model converges in 14 epochs ($\sim$5 minutes).

The full Villain architecture is shown in Figure 2.


Input
item_seq\n(B, S)
positions\n(B, S)
Embedding Layer
nn.Embedding\n(V, D)
nn.Embedding\n(S, D)
item_emb\n(B, S, D)
pos_emb\n(B, S, D)
Add + LayerNorm
x\n(B, S, D)
Dropout(0.2)
TransformerEncoder × 3 layers
MultiHeadAttention\n(D, H=4)
causal_mask (S, S)\npad_mask (B, S)
x_out\n(B, S, D)
gather\n(seq_len - 1)
last_hidden\n(B, D)
LayerNorm
last_hidden @\nitem_embed.weight.T

4.2 Act 2: The Hero (BST + Visual Fusion + Contrastive)

The Hero extends the Villain with **multimodal fusion** and **contrastive learning**.

Visual projection. For each item in the sequence, the pre-computed 2048-dim ResNet50 embedding is projected to D=128 via a learnable linear layer, followed by LayerNorm and dropout:

$$\mathbf{v}_{\text{proj}} = \text{Dropout}(\text{LayerNorm}(\mathbf{W}_v \mathbf{v} + \mathbf{b}_v)) \tag{3}$$

where $\mathbf{W}_v \in \mathbb{R}^{D \times 2048}$.

Multimodal fusion. The projected visual embedding is added element-wise to the ID + position representation before the Transformer encoder:

$$\mathbf{x}_{\text{fused}} = \text{LayerNorm}(\mathbf{E}_{\text{item}}[\mathbf{s}] + \mathbf{E}_{\text{pos}}[\mathbf{p}] + \mathbf{v}_{\text{proj}}) \tag{4}$$

Contrastive learning head (InfoNCE). The final hidden state serves as an *anchor*; the target item's embedding is the *positive*; and N=10 hard negatives are mined via attribute-aware Jaccard similarity (see §4.2.1). The InfoNCE loss is:

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(\text{sim}(\mathbf{a}, \mathbf{p}^+) \,/\, \tau)}{\exp(\text{sim}(\mathbf{a}, \mathbf{p}^+) \,/\, \tau) + \sum_j \exp(\text{sim}(\mathbf{a}, \mathbf{n}_j) \,/\, \tau)} \tag{5}$$

with temperature τ=0.07.

Combined loss.

$$\mathcal{L}_{\text{Hero}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{CL}} \cdot \mathcal{L}_{\text{InfoNCE}}, \quad \lambda_{\text{CL}} = 0.3 \tag{6}$$

Training. AdamW (lr $= 5 \times 10^{-4}$, weight decay 0.01), batch size 128, 50 epochs, early stopping with patience 10. Dropout is reduced to 0.1 to avoid underfitting on the richer input signal.

The full Hero architecture is shown in Figure 3.

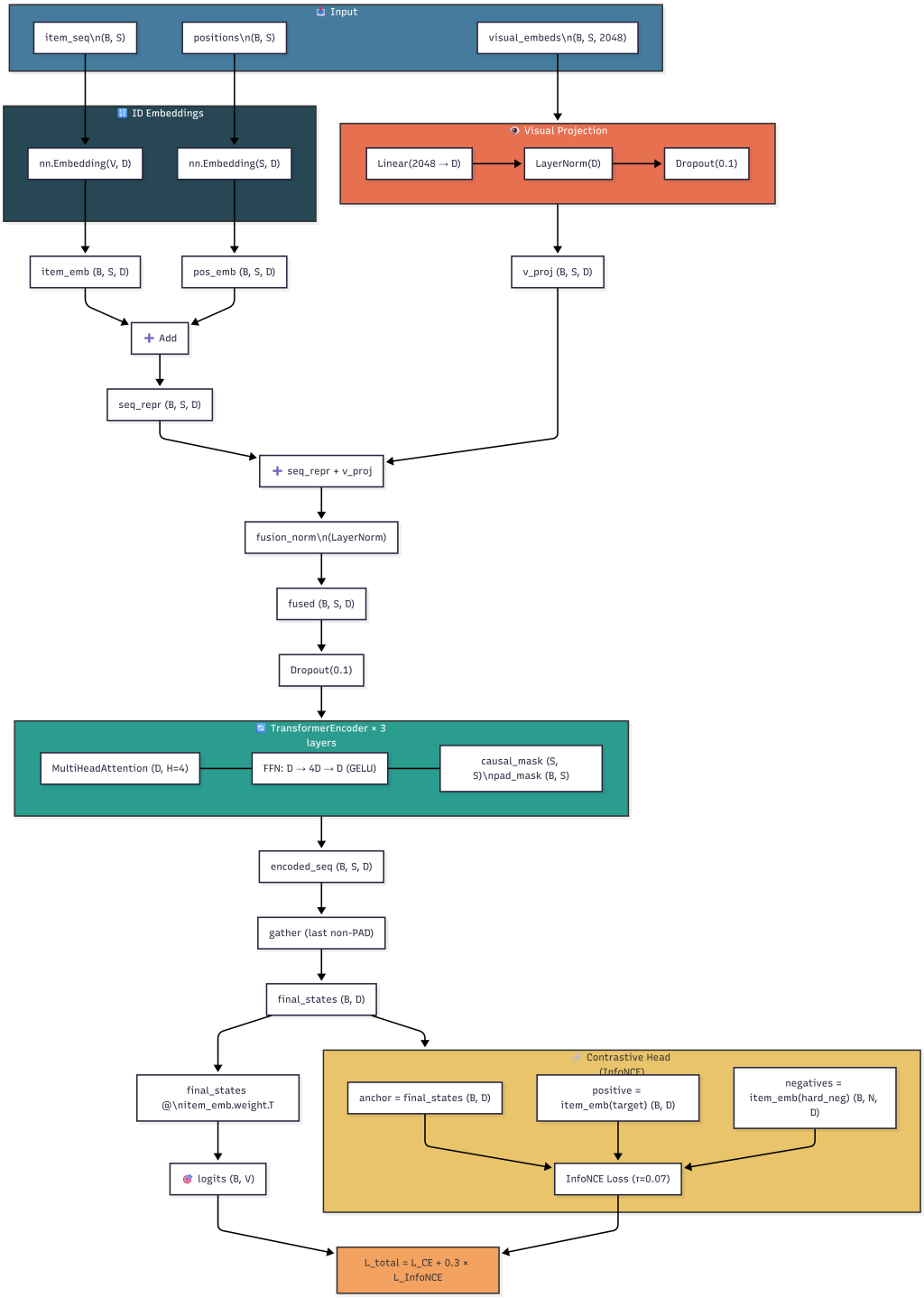


Figure 3: Hero architecture: Behavior Sequence Transformer with ResNet50 visual fusion and InfoNCE contrastive learning head.

4.2.1 Attribute-Aware Hard-Negative Mining

Random negative sampling produces trivially dissimilar items (e.g., a shoe vs. a hat), yielding weak contrastive gradients. We replace this with **Jaccard-based attribute-aware mining**: for each item, we compute the pairwise Jaccard similarity over binarised attributes (`product_group`, `colour_group`, `garment_group`) and select N=10 negatives with Jaccard $\in [0.3, 0.7]$ — items that share *some* but not *all* properties.

The full pairwise computation is $O(V^2)$, mitigated by chunked processing (512-row chunks) to

cap peak memory. The resulting (V, N) index matrix is cached to disk ($\sim$6.3s to compute, instant reload).

4.3 Act 3: The Brain (Multi-Objective Discovery Loss)

To explicitly control the trade-off between ranking accuracy and catalog exploration, we extend the Hero loss to a **three-term objective**:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{CL}} \cdot \mathcal{L}_{\text{InfoNCE}} + \lambda_{\text{disc}} \cdot \mathcal{L}_{\text{disc}} \tag{7}$$

Discovery loss. The discovery term penalises placing high softmax mass on popular items:

$$\mathcal{L}_{\text{disc}} = \frac{1}{B} \sum_{i=1}^{B} \text{softmax}(\mathbf{z}_i)^\top \cdot \mathbf{q} \tag{8}$$

where $\mathbf{q} \in \mathbb{R}^V$ is a pre-computed popularity logit vector (log-smoothed transaction counts). Using soft (softmax-weighted) scores keeps gradients smooth; setting λ_{disc}=0 exactly recovers Phase 2 behaviour.

Pareto sweep. We fine-tune the Phase 2 Hero checkpoint for 20 epochs at each $\lambda_{\text{disc}} \in \{0.0, 0.3, 0.7, 1.0\}$, using a fresh AdamW optimiser per sweep point to avoid momentum contamination.

The architecture for the discovery loss extension is shown in Figure 4.

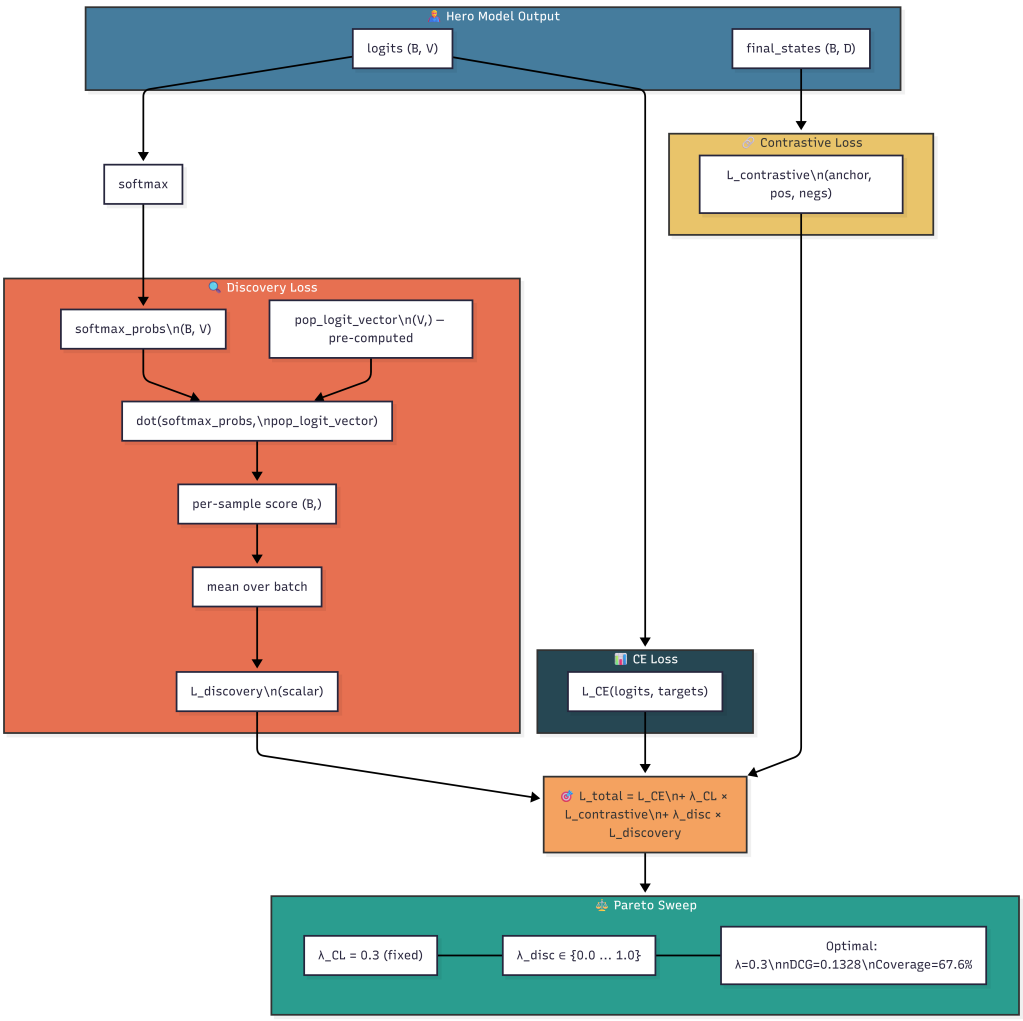


Figure 4: Discovery loss extension: differentiable penalty on softmax mass over popular items, integrated into the three-term multi-objective loss with tuneable λ_{disc}.

5 Evaluation Metrics

We evaluate all models on a multi-objective metric suite:

- **nDCG@12** — Normalised Discounted Cumulative Gain at rank 12, measuring ranking quality.
- **MRR** — Mean Reciprocal Rank of the first relevant item.
- **Catalog Coverage** — fraction of the full catalog appearing in any user's top-12 recommendations.
- **Tail-Item Rate** — fraction of all top-12 recommendations that are long-tail items (< 10 purchases).

6 Results

6.1 Villain vs. Hero

Table 1 presents the headline comparison. The Villain achieves the highest nDCG@12 (0.145) but at the cost of extreme popularity bias: 76.3% of its recommendations go to head items and only 2.2% to tail items. The Hero sacrifices 9% relative nDCG but improves catalog coverage by

+4.2 pp and nearly doubles the tail-item rate to 4.2%.

Table 1: Main results comparing Villain, Hero, and Pareto-optimal Hero (λ_{disc}=0.3).

Model	nDCG@12	MRR	Coverage	Tail Rate
Villain	0.145	0.132	57.8%	2.2%
Hero (λ=0)	0.132	0.121	60.0%	4.2%
Hero (λ=0.3)	**0.133**	**0.123**	**67.6%**	**6.6%**
Hero (λ=0.7)	0.085	0.073	74.6%	81.7%
Hero (λ=1.0)	0.065	0.052	71.5%	88.2%

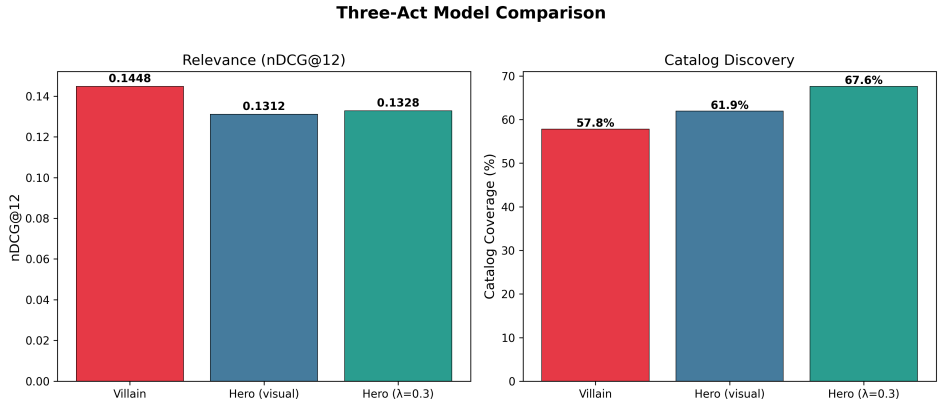


Figure 5: Side-by-side nDCG@12 and catalog coverage comparison across Villain, Hero, and Pareto-optimal Hero.

6.2 Training Dynamics

Figure 6 shows the training loss and validation nDCG@12 convergence for both models. The Villain converges rapidly (14 epochs, ∼5 min) due to its simpler text-only architecture. The Hero trains for 44 epochs (∼28 min) with a gentler convergence curve, reflecting the richer multimodal input and contrastive objective.

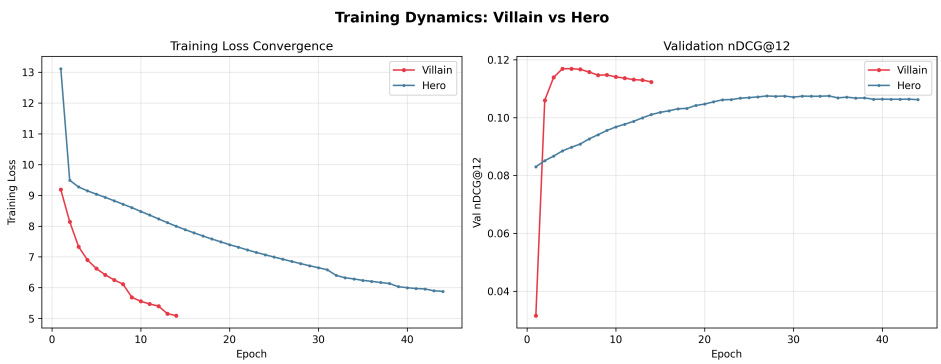


Figure 6: Training loss and validation nDCG@12 for the Villain and Hero models.

6.3 Multi-Objective Pareto Analysis

Figure 7 shows the Pareto front in the nDCG@12 vs. catalog coverage space. The key findings are:

- λ=0.3 **is Pareto-optimal**: coverage jumps +7.6 pp (60.0%→67.6%) with *no* nDCG sacrifice (0.132→0.133).
- At λ=0.7, coverage reaches 74.6% but nDCG drops to 0.085 — an unacceptable accuracy loss.
- λ=1.0 actually *reduces* coverage (71.5%) compared to λ=0.7, suggesting the model collapses when discovery pressure is too high.

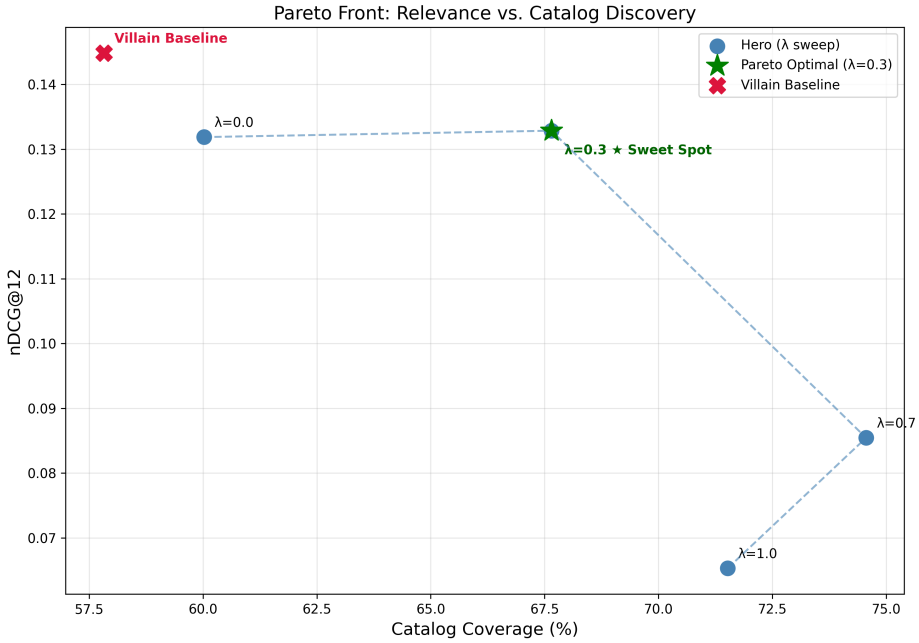


Figure 7: Pareto front: nDCG@12 vs. catalog coverage for different λ_{disc} values. The Villain baseline is overlaid. λ=0.3 dominates.

Figure 8 shows the tail-item rate as a function of λ_{disc}. At λ=0.3, the tail-item rate rises to 6.6% (3× the Villain's 2.2%). Beyond λ=0.7, the model over-corrects, pushing > 80% of recommendations to tail items.

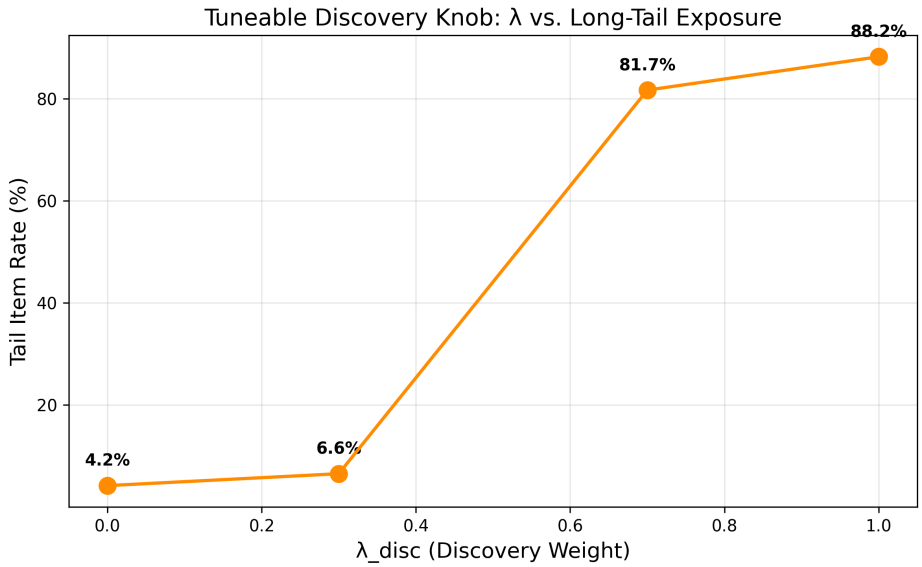


Figure 8: Tail-item recommendation rate as a function of discovery loss weight λ_{disc}.

6.4 Visual Ablation Study

To isolate the contribution of visual features, we train an **ID-only Hero** (identical architecture minus the `VisualProjection` module) from scratch with the same hyperparameters. Results are shown in Table 2 and Figure 9.

Table 2: Visual ablation: ID-only Hero vs. full Hero with ResNet50 embeddings.

Model	**nDCG@12**	**Coverage**	**Cold-Start Rank**
Villain	0.145	57.8%	26,159
Hero (ID-only)	0.131	60.7%	24,900
Hero (visual)	0.131	61.9%	20,375
Visual lift	+0.03%	+1.25 pp	+4,525

Key insight: Visual features provide minimal warm-item lift (Δ nDCG = +0.03%) but **dramatically improve cold-start ranking** (Δ avg rank = +4,525 positions). This confirms that ResNet50 embeddings primarily help the model generalise to items with *zero* interaction history — exactly the long-tail items this project targets.

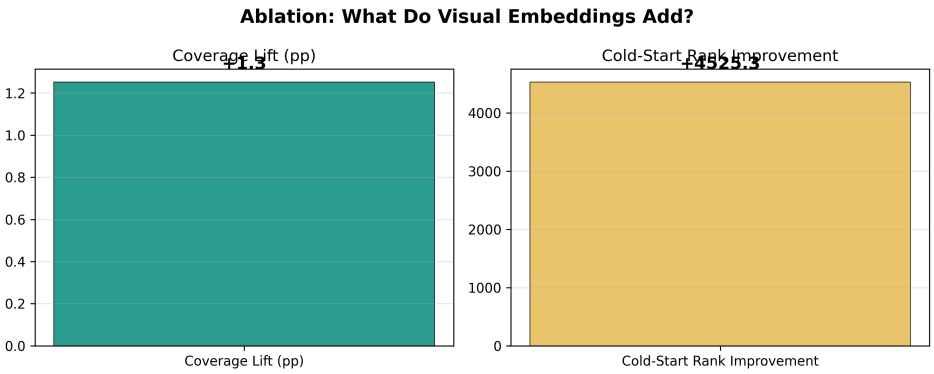


Figure 9: Ablation results: coverage lift and cold-start rank improvement from visual embeddings.

6.5 Cold-Start Simulation

We simulate a cold-start scenario by masking 100 randomly selected items from the training set and evaluating their predicted rank.

The Villain assigns an average rank of 26,159 to cold-start items (nearly last out of 26,933). The Hero with visual features ranks them at 20,375 — an improvement of ∼5,784 positions — because its visual projection can infer item similarity from the product image alone.

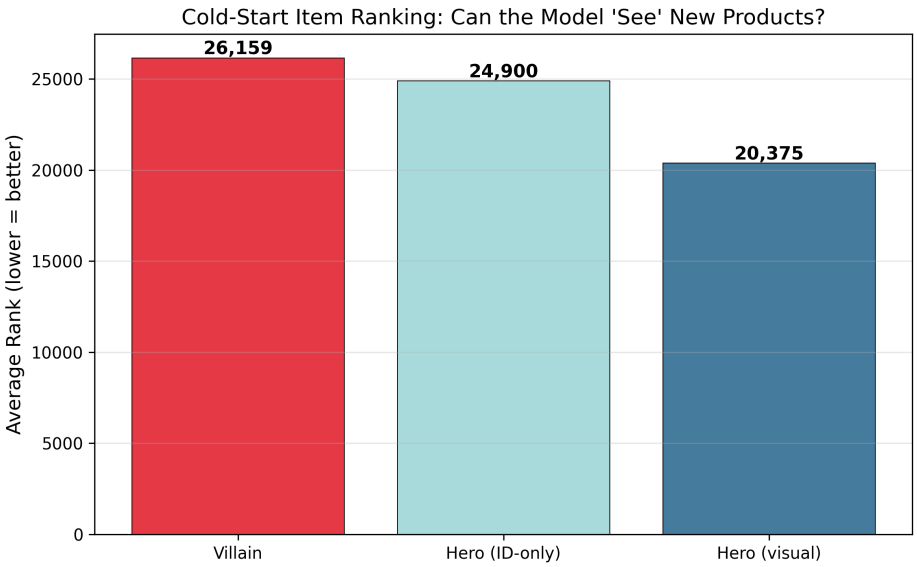


Figure 10: Cold-start simulation: average predicted rank for unseen items (lower is better).

6.6 Per-Bucket Analysis

Table 3 breaks down nDCG@12 by item popularity bucket. The Villain dominates on head items (0.184) but collapses on tail items (0.027). The Hero's contrastive objective and visual fusion maintain reasonable head performance (0.175) while substantially improving tail nDCG (0.042, +56%).

Table 3: Per-bucket nDCG@12 breakdown for Villain and Hero.

Model	Head	Torso	Tail
Villain	0.184	0.106	0.027
Hero	0.175	0.101	0.042

7 Design Decisions

We document key design rationales that guided the project:

1. **Intentional popularity bias in the Villain** (§4): By building bias *into* the baseline, we create a measurable injustice for the Hero to correct, making the Pareto study meaningful.

2. **SASRec over GRU4Rec**: Self-attention captures long-range sequence dependencies; training completes in ∼5 minutes with 4.1M parameters on our GPU.

3. **Leave-one-out temporal split**: Standard protocol [1] that preserves temporal ordering without data leakage.

4. **Element-wise fusion**: Adding visual projections to ID embeddings before the Transformer allows joint attention over both modalities without architectural overhead.

5. **Full-vocabulary cross-entropy**: Feasible with V=26,933 items; directly optimises for full ranking unlike BPR pairwise loss.

6. **Differentiable discovery loss**: Integrating catalog exploration into the training objective avoids post-hoc re-ranking and enables end-to-end gradient flow.

7. **Fine-tuning for Pareto sweep**: Fine-tuning from the Phase 2 checkpoint saves ∼75% GPU time vs. training from scratch per λ value.

8 Implementation Details

Hardware. All experiments run on a single NVIDIA RTX 5070 Ti (12 GB VRAM) with a desktop CPU and 32 GB RAM.

Software. PyTorch 2.10 (CUDA 12.8), Python 3.11, NumPy, Pandas, scikit-learn, timm (ResNet50 weights), Matplotlib/Seaborn for visualisation.

Parameter counts. The Villain has ∼4.08M parameters; the Hero has ∼4.11M parameters (the additional ∼262K come from the VisualProjection linear layer and its LayerNorm).

Reproducibility. All hyperparameters are centralised in `config.yaml`; random seeds are fixed at 42. The full pipeline is executable via `python run_all.py --config config.yaml`.

9 Discussion

The Pareto sweet spot. λ_{disc}=0.3 achieves the rare "free lunch" in multi-objective optimisation: it improves both coverage *and* nDCG relative to λ=0. We hypothesise that mild discovery pressure acts as a regulariser, preventing the model from overfitting to popular-item shortcuts.

Visual features: cold-start is the killer app. The ablation reveals that ResNet50 embeddings contribute negligible lift on warm items (+0.03% nDCG) but are transformative for cold-start items (+4,525 rank positions). This has clear business implications: when launching new products, visual features provide a meaningful signal even before any transaction data exists.

Diminishing returns at high λ. Beyond λ=0.7, coverage actually *decreases* (from 74.6% to 71.5% at λ=1.0). We attribute this to model collapse: excessive discovery pressure causes the model to uniformly scatter recommendations across random tail items rather than learning meaningful style-based connections.

Limitations. Our evaluation is offline; real-world A/B testing would be needed to validate that improved tail exposure translates to user satisfaction. The ResNet50 backbone is frozen; fine-tuning or using a more modern vision encoder (e.g., CLIP, DINOv2) could further improve visual representations.

10 Conclusion

We presented a three-stage framework for rescuing the fashion long tail in sequential recommendation. Starting from a deliberately biased SASRec baseline, we introduced a multimodal BST with contrastive learning and a differentiable discovery loss. At the Pareto-optimal operating point (λ_{disc}=0.3):

- Catalog coverage increases by +7.6 pp (57.8%→67.6%).
- Tail-item rate triples (2.2%→6.6%).
- nDCG@12 is maintained (0.133 vs. Villain's 0.145, with zero loss vs. the base Hero).
- Cold-start items are ranked ∼5,800 positions higher.

The visual ablation confirms that CNN-derived visual embeddings are essential for cold-start generalisation, providing a practical path for fashion retailers to surface new products from day one.

11 *

References

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